BUG-549MH2 Fri, 13 Mar 2026 13:47:43 GMT

INCORRECT TOTAL CALCULATION WHEN TAX RATE IS NEGATIVE

🔥 CRITICAL | P1 | POST | 200 OK | PRODUCTION

POST POST /api/v1/calculate-total → 200 OK

EXPECTED VS ACTUAL RESPONSE

✓ EXPECTED

HTTP 200 OK with JSON containing the correct total price including tax.

✗ ACTUAL

HTTP 200 OK returned, but the total value in the JSON response is incorrect.

STEPS TO REPRODUCE

1. Send a POST request to /api/v1/calculate-total.
2. Provide price and tax values in the request body.
3. Execute the request.
4. Observe the total value in the response JSON.
5. Compare it with the expected calculated value.

REQUEST PAYLOAD / HEADERS

Content-Type: application/json

IMPACT ASSESSMENT

AFFECTED USERS	DATA RISK
ALL USERS	HIGH RISK
SERVICE UPTIME	**SLA STATUS**
DEGRADED	BREACHED

⬡ AI ROOT CAUSE ANALYSIS

The API returned an unexpected response. Based on the status code 200 and response body, the most likely cause is a runtime error in the request handler, possibly due to unvalidated input, a missing dependency, or an unguarded external service call. Review server logs around the timestamp of this report for the full stack trace.

⊙ AI FIX SUGGESTIONS

1. Add comprehensive request logging with a unique trace ID to correlate client errors with server logs.
2. Implement input validation middleware to reject malformed requests early with a 400 and a descriptive message.
3. Add health check and readiness probe endpoints to surface dependency failures before they cascade.
4. Write an integration test that reproduces this exact request to prevent regression.

Reporter: **Namaste.Sahana@Gmail.Com** Fri, 13 Mar 2026 13:47:43 GMT BUG-549MH2 BUG REPORT